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1. THE PAYMENT SYSTEM

In every economy, a large number of transactions take place each day on the initiative of a wide range of economic actors. So, almost all of these transactions involve transfer of money.

A **payment system** is a system used for transferring money between physical and legal persons done for paying some obligation.

The term “payment system” refers to the complete set of instruments, intermediaries, rules, procedures, processes and interbank funds transfer systems which facilitate the circulation of money in a country or currency area.

Well-designed payment infrastructure contributes to the proper functioning of markets and helps to eliminate frictions in trade.

In payment system transfer of money can be done like a cash or non-cash payments.

Also, with the advent of computers and electronic communications a large number of alternative electronic payment systems have emerged. These include debit cards, credit cards, electronic funds transfers, direct credits, direct debits, internet banking and e-commerce payment systems.

Payments can be:

1. Domestic payment and
2. International payment.

Domestic payments can be used to make transfers between physical and legal persons from the same country.

International payment are those which are used to make transfers between resident of different countries.

1.1. TYPE OF RISKS IN PAYMENT SYSTEM

In payment system, participants face the risks that could be divided in three categories:

1. credit risk,
2. liquidity risk and
3. systemic risk.

¹**Credit risk** is the risk that a counterparty will not settle an obligation for full value, neither when that obligation becomes due nor at any time thereafter.

¹ Kokola, T., *The payment system*, European Central Bank, Frankfurt, 2010.

Liquidity risk is the risk that a counterparty will not settle an obligation for full value when it becomes due. This does not imply that the counterparty or participant is insolvent, since it may be able to effect the required settlement at some unspecified time thereafter. Liquidity risk materialises if a party does not have the necessary funds or assets at its disposal when the obligation becomes due.

Systemic risk is the risk that the inability of one participant to discharge its obligations in a system will cause other participants to be unable to fulfil their obligations when they become due. This could potentially result in significant liquidity or credit problems spilling over into other systems or markets, thereby threatening the stability of the financial system. The original inability to discharge obligations may be caused by operational or financial problems.

In order to this risks will be at minimum level paying system must functioning effectively.

2.1. TYPES OF PAYMENT SYSTEM

A payment system is usually classified as a “large-value” or “retail” payment system depending on the main type of transaction processed in the system.

Large-value payment systems (LVPSs), also called “wholesale payment systems”, are systems which are designed primarily to process urgent or large-value payments.

Retail payment systems are designed to handle a large volume of relatively low-value payments, such as credit transfers, direct debits and card payments. Retail payment systems may settle in either central bank or commercial bank money.

Depending on their settlement methods, payment systems are divided into four design types, with the most common forms being real-time gross settlement and designated-time net settlement (DNS).

- Real-time gross settlement systems effect the final settlement of individual –payments on a continuous basis during the processing day and are the predominant form of LVPS.
- Designated-time net settlement systems settle the net positions of participants at –one or more discrete pre-specified settlement times during the processing day. This is the main form of retail payment system, often with several settlement cycles during the day.
- Net settlement LVPSs usually settle once at the end of the day. Designated-time gross settlement systems exist in some countries. In these – systems, the final settlement of transfers occurs at the end of the processing day with no netting of credit and debit positions – i.e. on a transaction-by-transaction basis or on the basis of the aggregate credit and debit positions of each bank.
- Hybrid systems combine the features of gross and net settlement – e.g. frequent – offsetting of transactions and frequent final settlement during the day.

3.1. RTGS (Real Time Gross Settlement)

RTGS system is a funds transfer mechanism where transfer of money takes place from one bank to another on a "real time" and on "gross" basis. names for it.

-Settlement in "real time" means that payment are settled as soon as they are processed, so there is no intraday exposure to credit risk.

- "Gross settlement" means the transaction is settled on one to one basis without bunching with any other transaction.

Payment instructions between banks are processed and settled individually, provided that the payer has sufficient funds on his account. This is the fastest possible money transfer system through the banking channel. It is online system and different countries have different names for it.

²RTGS systems are typically used by financial institutions for the settlement of large-value and/or time-critical payments – e.g. money market transactions, foreign exchange transactions and the cash leg of securities transactions. These systems are also used for the settlement of settlement obligations stemming from ancillary systems such as retail net settlement systems. As a result, transaction values in RTGS systems are usually very high. RTGS systems process credit transfers, which are initiated by the payer. These are settled by (simultaneously) debiting the payer's account and crediting the beneficiary's account, after which a payment is considered to be final. The number of RTGS systems increased dramatically in the 1990s, and today most modern economies have an RTGS system. In most RTGS systems, the settlement bank is the central bank, which typically also owns and operates the system.

3.1.1. Goals of RTGS

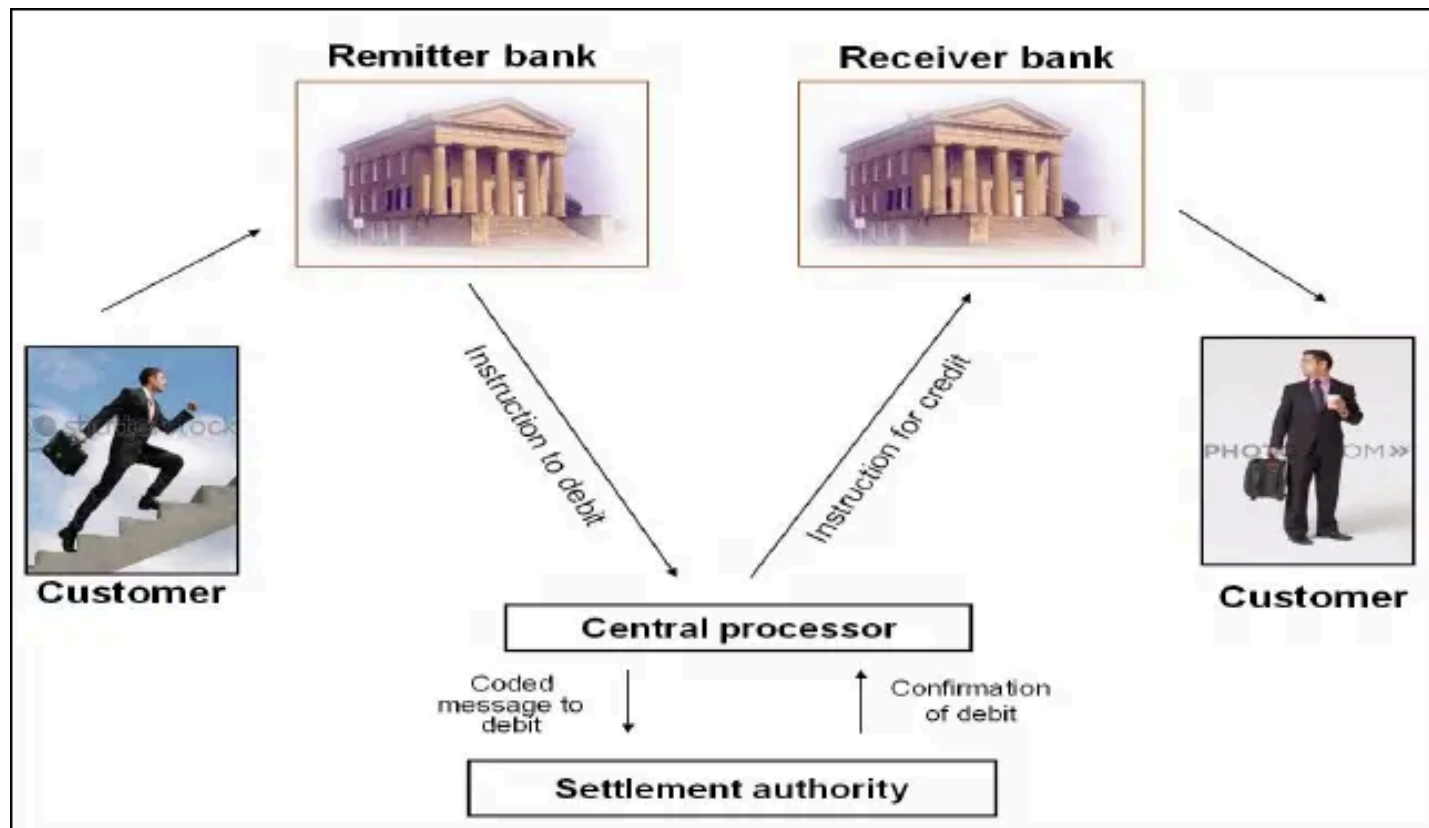
The goals of RTGS are:

1. To reduce settlement risk due to settlement lag
2. To reduce credit risk
3. Speed up the process of high value payments
4. To give accurate position of the participating bank

² Kokola, T., *The payment system*, European Central Bank, Frankfurt, 2010., page 53.

How it works?

The process begins with a customer who wants to make a money, so he makes a request to remitter bank. Remitter bank debits the account of the customer and sends instruction to central processor, to debit its account. Central processor prepares a coded message which debits account of remitter bank with the settlement authority. So, the message is sent to settlement authority to debit remitter bank's account. Then there is (from settlement authority again to central processor) a confirmation of debit of account of remitter bank's account.



Instructions for credit are sent to receiver bank, to credit account of the customer.

-If the sending bank does not have sufficient covering funds in its central bank account then we have two different situations:

1. One possible way is for the system to reject the orders and return them to the sending bank. The rejected transfer orders will be input into the system again at a later time when the sending bank has covering funds. Until that time, sending banks may keep and control the pending transfers within their internal systems.
2. Alternatively, the RTGS system may temporarily keep the transfer orders in its central processor, instead of rejecting them. In this case, the pending transfers will be released for settlement when covering funds become available on the basis of predefined rules, agreed between the system and the participating banks.

2. INSTRUMENTS OF PAYMENT

A payment system comprises three main elements or processes:

1. payment instruments, which are a means of authorising and submitting a payment (i.e. the means by which the payer gives its bank authorisation for funds to be transferred or the means by which the payee gives its bank instructions for funds to be collected from the payer);
2. processing (including clearing);
3. a means of settlement for the relevant banks.

A payment instrument is a tool or a set of procedures enabling the transfer of funds from the payer to the payee. There are a variety of different payment instruments, each with its own characteristics depending on the type of relationship and transaction between the payer and the payee. The most common distinction is between cash and non-cash payment instruments.

2.1. TYPES OF PAYMENT INSTRUMENTS

Cash payments (i.e. payments made using banknotes and coins) are usually associated with face-to-face transactions of low value between individuals or between an individual and a merchant. If the parties do not exchange information on their identity, a cash payment is said to be “anonymous”.

Non-cash payments, by contrast, involve the transfer of funds between accounts. A non-cash payment instrument is therefore the means by which a payer gives its bank authorisation for funds to be transferred or by which a payee gives its bank instructions for funds to be collected from a payer. The accounts of the two parties may be held with a single bank or with different banks.

Non-cash payment instruments can be further classified on the basis of the following:

- Physical form (paper-based or electronic instruments)
Payment instructions have traditionally been in paper form, but today they are increasingly taking the form of electronic instructions.
- The party submitting the payment instrument for processing (credit or debit-based instruments).

Credit-based (“credit push”) instruments are submitted for processing by the payer, while **debit-based** (“debit pull”) instruments are submitted for processing by the payee. The main credit-based instruments are credit transfers (also called “direct credits” or “wire transfers”). The main debit-based payment instruments are direct debits, card payments and cheques. In a credit-based transfer the instruction and funds move in the same direction, whereas in a debit-based transfer they move in opposite directions.

Electronic money (or “e-money”) is a monetary value represented as a claim on the issuer which is stored on an electronic device and accepted as a means of payment by undertakings

other than the issuer (by contrast with single-purpose prepaid instruments, where the issuer and acceptor are one and the same). E-money can be either hardware-based (i.e. stored on a device, typically a card) or software-based (i.e. stored on a computer server). E-money can be regarded as a means of settlement rather than a payment instrument, since the creation or reimbursement of e-money is effected using one of the core payment instruments – cash, payment cards, direct debits or credit transfers.

The most commonly used cashless payment instruments are payment cards, credit transfers, direct debits and cheques.

Credit transfers are the most widely used means of non-cash payment in the euro area, followed by direct debits, as these means of payment offer the most convenience to their users. Also on the rise are card-based payments, with debit cards being preferred to credit cards in most countries. Although traditionally a very important payment instrument, in many countries of the euro area cheques have been replaced to a large extent by other payment instruments. Even in countries where the actual number of cheque payments is still rising (Ireland, Italy), their importance relative to other payment instruments is declining. A further reduction in the use of cheques can be expected owing to the discontinuation of the eurocheque guarantee at the beginning of 2002.

Payment systems are used in lieu of tendering cash in domestic and international transactions and consist of a major service provided by banks and other financial institutions.

The difference is **internal payment** operations and **foreign payment** transactions, depending on whether the participants from one country or one of the non-resident participants. If participants are payments from one country talking about the internal payment system.

National payments can be viewed in the narrow and broad sense. Taken in a broader sense, includes all payments within one country, regardless of whether it is done through the appropriate institutions or not. This includes all payments which are mediated between private individuals, between individuals and legal entities, as well as the mutual payments of legal entities. On the other hand, socially organized payment covers only the cash payments between participants in the payments that are made through accounts which are maintained under the law authorized institutions - holders of payment in the country.

International (foreign) payments between residents of different countries with different currency, customs and legal systems, whereby one currency into another.

Risks in international business

- Commercial risks
- Currency risk
- Political Risk.

INTERESTS OF THE PARTIES

Payment risk: Export financing.

The risk of execution: Import financing.

INSTRUMENTS OF PAYMENT FOR INTERNATIONAL SALES:

- Check, bills of exchange, remittance, INKASO (D/C – documentary collection), D/C ili L/C – documentary credit ili letter of credit.

3. PAYING INSTRUMENTS

3.1. CHEQUE

Cheque is an instrument in writing containing an unconditional order, addressed to a banker, sign by the person who has deposited money with the banker, requiring him to pay on demand a certain sum of money only to or to the order of certain person or to the bearer of instrument.

3.1.1 TYPES OF CHEQUE

Commercial Checks/Personal Checks

Commercial or personal checks are written by individuals or businesses for varied amounts. Unless otherwise noted on the face of the check, these items become stale dated six months after issuance.

Traveler's Checks

Traveler's Checks are negotiable instruments purchased by the user for a fixed amount. Because they are pre-paid, these items do not expire or become stale dated.

Official Checks/Cashier's Checks

Official checks or cashier's checks are direct obligations of the bank on which they are drawn. These drafts, in which the drawer and drawee are the bank or branches of the same bank, are signed by an officer or employee of that bank.

3.1.2. ELEMENTS OF CHEQUE



1. Name of bank which is also called the "drawee bank" or paying bank
2. "Account Payee Only" crossing is a directive to the collecting bank to pay into the account of the payee
3. Payee is the person to whom the cheque is to be made. Ensure that the name of the person is correctly spelt and written close to the words "pay to". Draw a line on the space after the payee's name to avoid alteration
4. Date of the cheque. To be able to receive payment, the date must be the current date
5. The person who holds and presents the cheque at the bank. It is advisable to cross out "or bearer" to avoid any stolen cheque from being paid out
6. The payment amount written in words. The same value will be written in the box beside it. Ensure that the amount in words and figures are written close to the words "Ringgit Malaysia" and "RM" printed on the cheque. Do not leave any gap by drawing a line after them. Make sure that the amount in words and in figures tally, otherwise, the bank will return the cheque
7. Your signature as the "drawer" of the cheque
8. Serial number of the cheque. Each cheque has a different number for identification purposes
9. The drawee bank's state and branch code
10. Your current account number
11. The drawee bank's internal code for current account product

Until recently, consumers used checks more often than any other retail payment instrument in the United States other than cash. Checks are very convenient payment instruments. Consumers can use them at the point of sale, for bill payments, and for person-to-person transactions. Nonetheless, checks comprise a decreasing percentage of the total non cash payment volume in the United States.

4.1. PAYMENTS CARD

Over the last ten years payment cards have displayed the strongest growth, with transaction volumes for this instrument. Consequently, payment cards have overtaken credit transfers as the most widely used non-cash payment instrument in the euro area. *Payment cards* are used almost exclusively for consumer purchases of relatively low value.

Financial institutions are important participants in various credit card systems. They issue and distribute cards, clear and settle the associated payments, and in some cases act as merchant acquirers.

Payment cards allow its holder to buy goods and services based on the holder's promise to pay for these goods and services. The issuer of the card creates a revolving account and grants a line of credit to the consumer (or the user) from which the user can borrow money for payment to a merchant or as a cash advance to the user.

Credit cards

Credit card, in today's sense of the word, is a card that proves that its owner is granted a credit line. It allows the owner to purchase and / or raise cash to pre-determined amount, approved a loan can be paid in full at the end of a period or may be paid in installments, with the outstanding debt is considered an approved loan. Interest is charged on the amount of any approved loan, but the owner is often charged an annual fee to maintain the card or account.

Debit cards

Unlike a credit card, debit card is a card that allows its owner to buy direct debit your account. Debit cards allow you to pay and raise the cash to the extent you have in your account, including minus allowed in some banks. In many countries the use of debit cards has become so widespread that their volume of use has overtaken or entirely replaced the check and, in some instances, cash transactions

Prepaid cards

The definition of a prepaid credit card is: A payment card (usually MasterCard, Maestro, Visa Electron or Amex), pre-loaded with your own money, which you can then use wherever the payment card is accepted, including on the internet and abroad.

Digital money

Electronic money and electronic payment is the exchange of material goods through the telecommunication infrastructure, such as intranet systems of banks or the Internet. This money is basically virtual and presented a numerical system that exists in computer memory, and as such knows no geographic boundaries and can be convenient at the time of transfer over long distances. Although often used the term "electronic money", more precise terminology is called "digital money", because the first can be used in analog communications.

Electronic money allows purchase of goods and services through computers in the commercial computer networks (eg Internet) or business banking networks (eg SWIFT).

The dominant form of electronic money is the electronic transfer of funds at point of sale (EFT / POS) with terminals installed in commercial and service network. Another form of use of electronic money is available through bankomats that allow cash withdrawals.

5.1. LETTER OF CREDIT

Letters of credit and bills of exchange are sometimes also referred to as payment instruments, although they are usually credit instruments which can, in some circumstances, be used for payments.

Letter of credit (LC) - An irrevocable commitment by a bank (the issuing bank) or other issuer made at the request of a customer (the applicant third party) to pay a specified sum of money to a third party upon request, subject to terms and conditions drawn up in accordance with uniform customs and practices. It is frequently used in international trade to make funds available in a foreign location.

A bill of exchange is a written order from one party (the drawer) to another (the drawee) to pay a specified sum to the drawer (or a third party specified by the drawer) on demand or on a specified date. These are widely used to finance trade and, when discounted with a financial institution, to obtain credit.

5.1.1 ELEMENTS OF A LETTER OF CREDIT

- A payment undertaking given by a bank (issuing bank)
- On behalf of a buyer (applicant)
- To pay a seller (beneficiary) for a given amount of money
- On presentation of specified documents representing the supply of goods
- Within specified time limits
- Documents must conform to terms and conditions set out in the letter of credit
- Documents to be presented at a specified place

6.1.1. TYPES OF LETTER OF CREDIT

Revocable letter of credit

Just like the name says the LC can be revoked by the Issuing Bank without the agreement of the beneficiary.

Irrevocable letter of credit

Can not be cancelled or amended without all the parties agreement.

Standby letter of credit

Guarantee of payment. If the beneficiary does not get paid from its customer it can then demand payment from the Bank by forwarding the copy of the invoice that was not paid and supporting documentation.

Revolving letter of credit

It is established when there are regular shipments of the same commodity between supplier and customer. Eliminates the need to issue an LC for each individual transaction.

Example:

For simplicity sake let's imagine that your company imports radios from a Korean manufacturer called **Seoul Manufacturing**, which banks at **First Seoul Bank**. Your company currently banks at **First American Bank**

For the purpose of this example these will be the roles that the parties will play in the letter of credit transaction:

Your company : applicant

Seoul Manufacturing : beneficiary

First American Bank : Issuing Bank

First Seoul Bank : Advising Bank

The example: You want to buy \$50,000 worth of radios from Seoul Manufacturing, which agrees to sell the merchandise and gives you 60 days to pay it with the condition that you provide them with a 90 days letter of credit for the full amount. The steps to get the LC would be as follows:

- 1)** You go to First American Bank and request a \$50,000 letter of credit with Seoul Manufacturing as a beneficiary.
- 2)** The bank goes through its underwriting process. Although the bank is not advancing money, they are extending credit on your behalf and are taking on a contingent liability. If your company qualifies from a credit standpoint the LC is issued.
- 3)** Even if your company does not qualify for credit, you can still get an LC if you are willing to put cash collateral. CD secured letters of credit are very common for small business.
- 4)** The bank sends a copy of the letter of credit to First Seoul Bank, which lets the vendor know and the merchandise is shipped.

Take into consideration that the letter of credit itself might be the source of repayment of the transaction. It could be that Seoul Manufacturing is interested in getting paid as soon as the merchandise is shipped. Therefore, the letter of credit will indicate that payment shall be made as soon as Seoul Manufacturing can present proof of shipping.

If the letter of credit that your vendor requires is not tied to a particular transactions, but they are asking for a guarantee that makes sure that you will not default. They are probably asking for a **Stand-By letter of credit** or a **Revolving letter of credit**. These types of LCs are usually for a longer term. Usually a year and are the vendor's guarantee that they will get paid.

The example above describes the simplest of letter of credit transactions. Although there are other factors involved such as the role of correspondent banks and confirmations, the thing that you should be concerned as a customer is expediency and the fees involved, which can run anywhere from 1.5% to 8% of the value of the LC.

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